Standards-Based Validation of LV Endocardial Border Tracing

Comparison of the purposed echocardiographic contours with the 2015ASE/EACVI chamber-quantification recommendations (Lang et al.)

ASE → American Society of Echocardiography

- EACVI → European Association of Cardiovascular Imaging

Purpose. To determine whether the boundary traced by the proposed algorithm corresponds to the clinically relevant LV cavity boundary described by the ASE/EACVI guideline, and to document the comparison in a reproducible report.

Bottom-line finding. On visual inspection, all 7 submitted examples place the yellow contour along the apparent blood–tissue interface of the LV cavity, with the myocardium lying outside the traced region. This is consistent with the boundary used for 2D LV volumetric tracing in Lang et al. 2015.

Important scope statement. This report establishes standards consistency of the displayed boundary. It is not a clinical validation study and cannot establish diagnostic accuracy, inter-observer agreement, Dice/IoU accuracy against a reference contour, or suitability for clinical deployment without an independent reference standard.

Primary source

Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for Cardiac Chamber Quantification by Echocardiography in Adults: An Update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. J Am Soc Echocardiogr. 2015;28(1):1–39.e14. doi:10.1016/j.echo.2014.10.003.

The same joint guideline was published in European Heart Journal – Cardiovascular Imaging 2015;16(3):233–270. doi:10.1093/ehjci/jev014.

1. What the ASE/EACVI guideline actually requires

The key statement is in thesection on LV size/volume measurement. Langet al. statethat volumetric measurements are based on tracing the **blood–tissue interface** in apical four- and two-chamber views. They further specify that the contour is closed at the mitral-valve level by connecting the opposite sections of the mitral ring.

> **Why this establishes the relevant boundary:** the blood–tissue interface is the interface separating the LV blood-filled cavity from the myocardial wall. Therefore, for a cavity-area/volume contour, the clinically relevant boundary is the **endocardial border** rather than the epicardial outer wall.

The guideline also explicitly discusses **endocardial border tracings**, stating that image acquisition should minimize errors in these tracings. It recommends contrast when needed to improve endocardial delineation when two or more contiguous LV endocardial segments are poorly visualized.

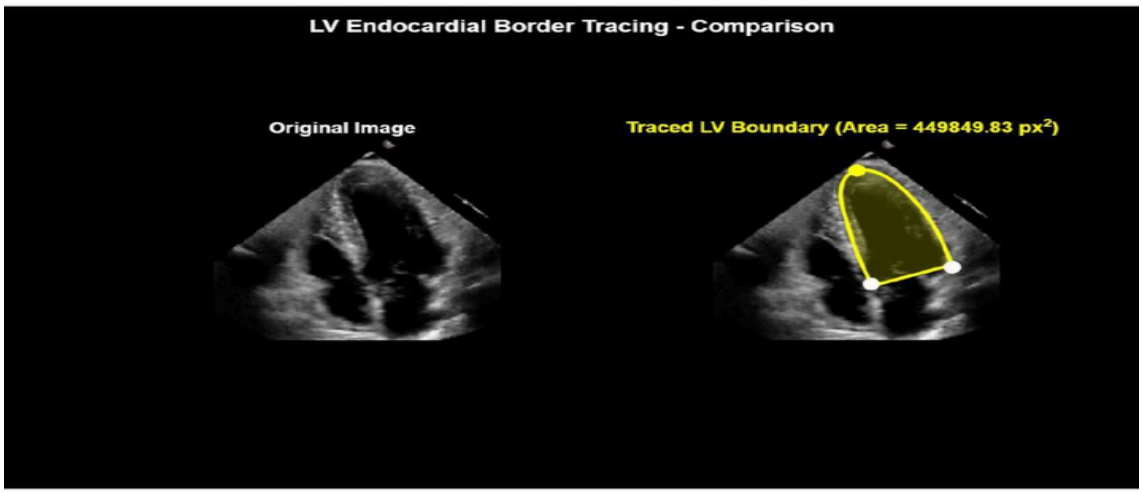
Direct guideline evidence (source location): Lang et al., JASE 2015, pp. 3–5; official ASE-hosted PDF pp. 3–5. The document says: (1) volumetric measurements are based on the blood–tissue interface; (2) the contour is closed at the mitral valve level; and (3) image acquisition should minimize errors in endocardial border tracings.
Source: American Society of Echocardiography, official copy of Lang et al. 2015. DOI 10.1016/j.echo.2014.10.003.

2. Translation into a validation rule for your algorithm

Criterion	ASE/EACVI basis	What is checked in your images
1. Target is LV cavity	Volumetric tracing follows the blood–tissue interface.	Yellow region is inside the LV cavity rather than around the outer myocardial wall.
2. Boundary is endocardial	The blood–tissue interface corresponds to the endocardial border.	Yellow line follows the visible inner cavity edge; myocardial tissue remains outside the region.
3. Apical contour	The LV contour extends to the apical point/most distant point of the LV contour.	The yellow contour reaches the apparent LV apex in each example.
4. Basal closure	At the mitral-valve level, the contour is closed by a straight line connecting the opposite mitral-ring sections.	The basal ends are connected by a straight/near-straight closing segment.
5. Avoid myocardial inclusion	The purpose is cavity measurement, not epicardial measurement.	The displayed filled region generally excludes the surrounding myocardial wall.

Important distinction: the guideline is a measurement standard, not an algorithm specification. It does not require three manual points, an ellipse, PCHIP interpolation, four draggable handles, FRFCM, or any particular segmentation method. Those are implementation choices. The medically relevant question for this document is whether the final contour represents the correct LV cavity/endocardial boundary.

3.1 Image-by-image comparison — submitted example 1



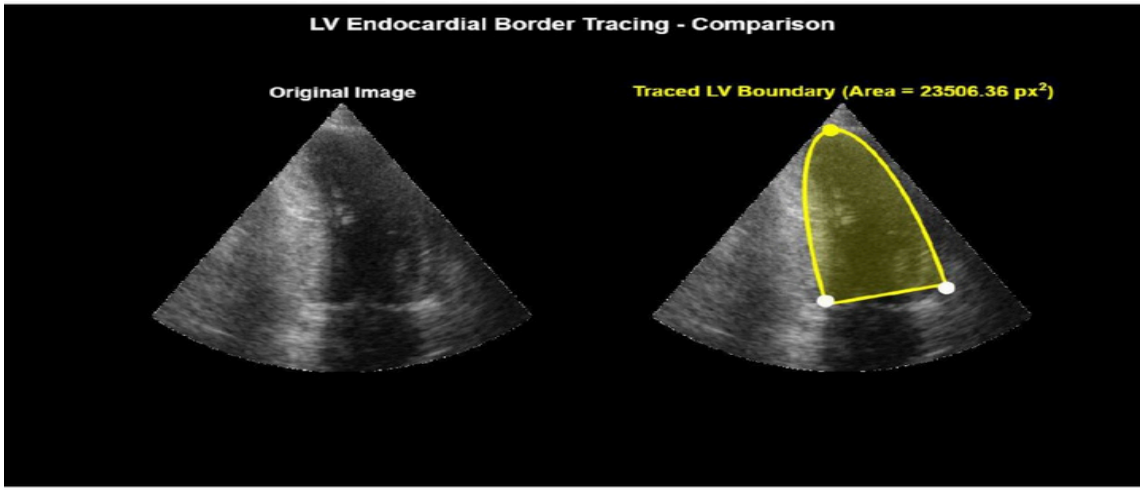
Displayed area: 44,984.83 px^2

Standards-consistency assessment: Consistent with the endocardial-border criterion.

Visual reasoning: The yellow contour is placed within the LV cavity and follows the inner cavity edge from the basal region toward the apex. The filled area represents the cavity rather than the outer myocardial boundary. The basal closure is drawn between the two basal endpoints, consistent with the guideline's description of closing the contour at the mitral-valve level.

This is a visual comparison against the anatomical/measurement criterion in Lang et al.; it is not a claim of expert-annotated ground-truth accuracy.

3.2 Image-by-image comparison — submitted example 2

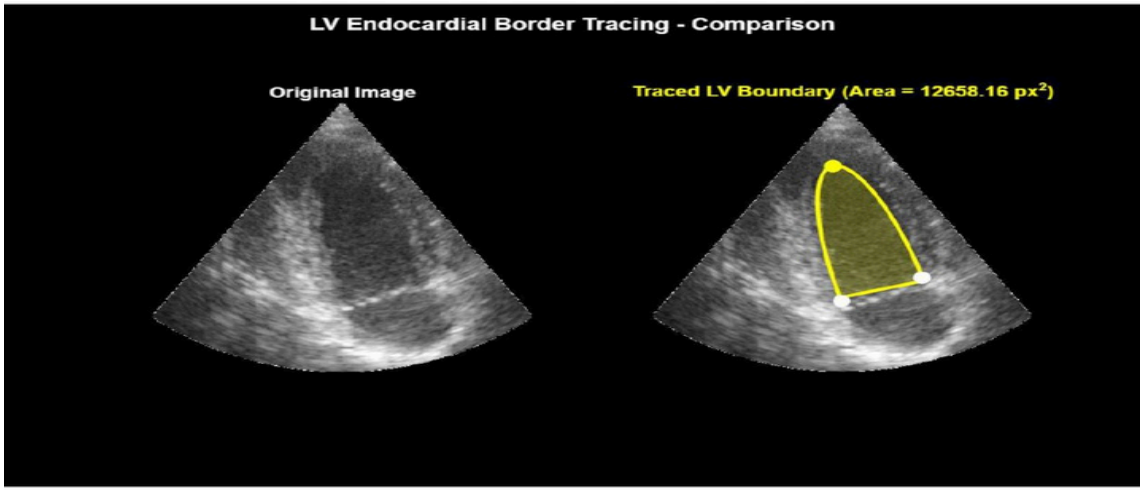


Displayed area: 23,506.36 px^2

Standards-consistency assessment: Consistent with the endocardial-border criterion.

Visual reasoning: The contour tracks the dark blood-pool region and remains inside the myocardial wall. The apex is included, and the basal line closes the cavity region. No obvious epicardial/outside-wall tracing is used.

3.3 Image-by-image comparison — submitted example 3

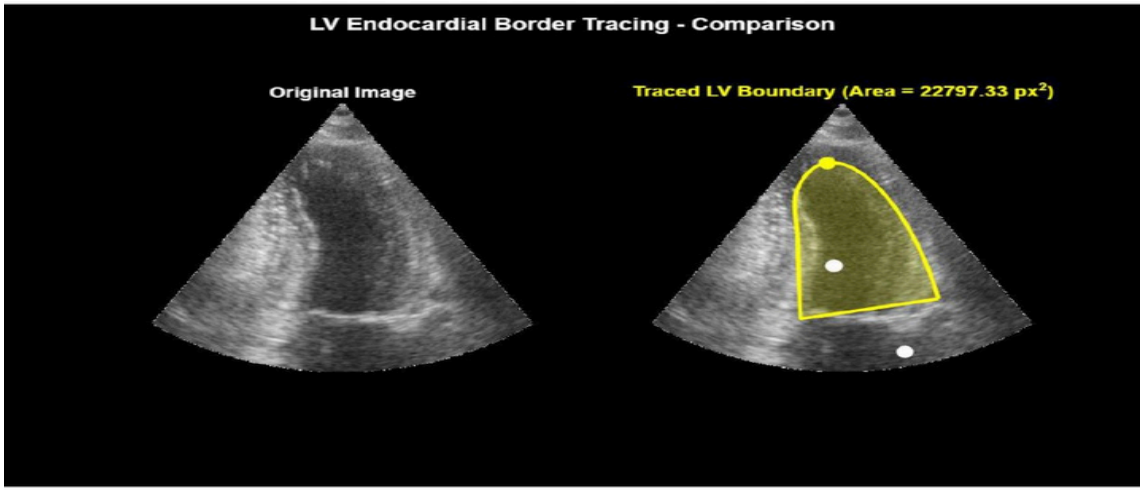


Displayed area: 12,658.16 px^2

Standards-consistency assessment: Consistent with the endocardial-border criterion.

Visual reasoning: The traced region follows the apparent inner LV cavity. The myocardial wall is outside the yellow boundary. The contour extends toward the apex and closes basally.

3.4 Image-by-image comparison — submitted example 4

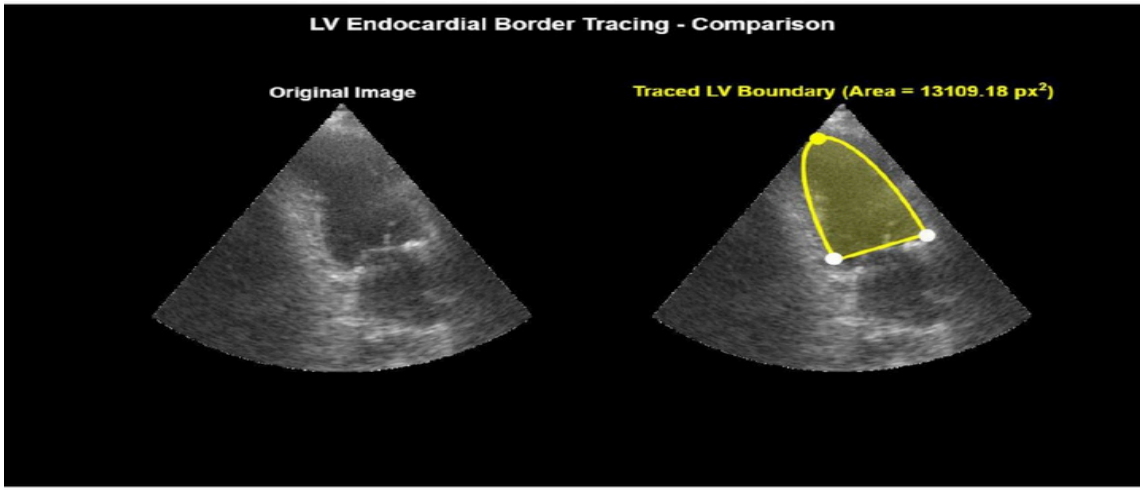


Displayed area: 22,797.33 px^2

Standards-consistency assessment: Consistent with the endocardial-border criterion, with a basal-placement caveat.

Visual reasoning: The contour is positioned along the apparent inner cavity boundary and excludes the surrounding wall. The basal closure is visible. The image quality makes the exact basal endocardial edge less sharply defined than in an ideal study, so this example should be treated as visually consistent rather than quantitatively proven.

3.5 Image-by-image comparison — submitted example 5

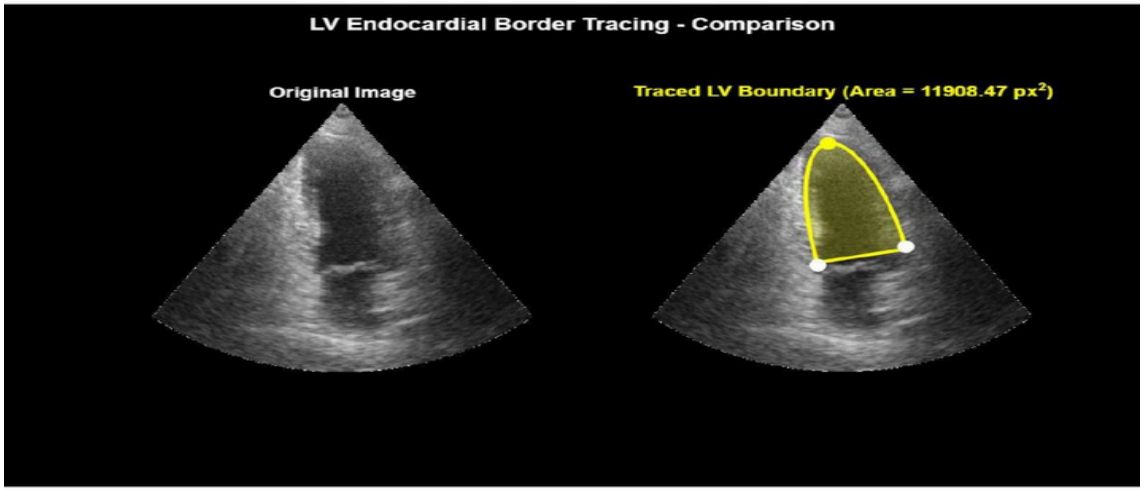


Displayed area: 13,109.18 px^2

Standards-consistency assessment: Broadly consistent, but lower-confidence because of image quality.

Visual reasoning: The traced area follows the visible cavity rather than the outer wall and reaches the apical region. However, the source image is relatively low-contrast/noisy, so the precise endocardial interface is less certain by visual inspection.

3.6 Image-by-image comparison — submitted example 6

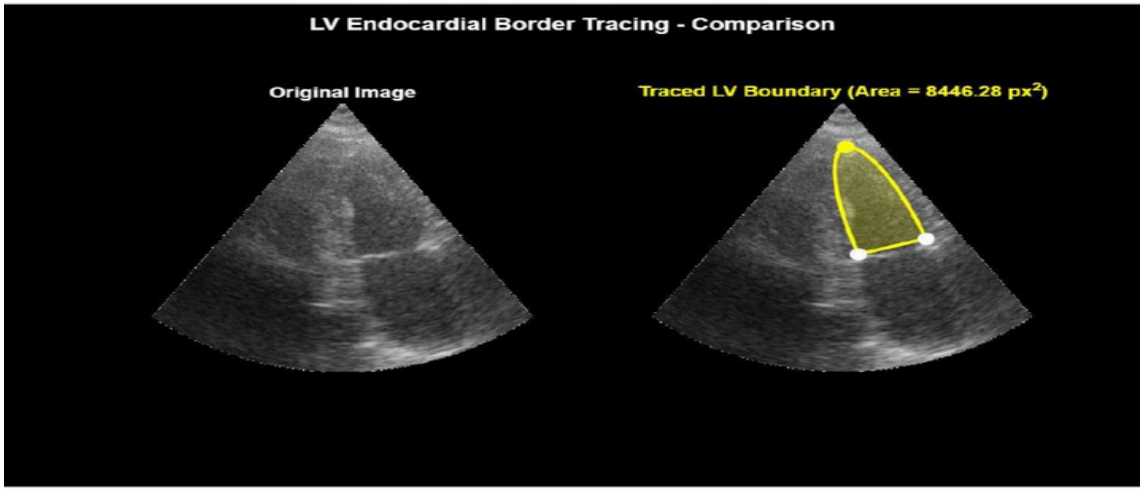


Displayed area: 11,908.47 px^2

Standards-consistency assessment: Consistent with the endocardial-border criterion.

Visual reasoning: The yellow line follows the apparent inner cavity border, with myocardium outside the region. The basal closure and apical extent are compatible with the guideline's LV contour description.

3.7 Image-by-image comparison — submitted example 7



Displayed area: 8,446.28 px^2

Standards-consistency assessment: Broadly consistent, but lower-confidence because of image quality

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Visual reasoning: The contour is placed within the apparent LV cavity and does not trace the outer myocardial wall. Because the underlying image has limited border contrast, the exact location of the endocardial interface cannot be established with the same confidence as a high-quality image.

This is a visual comparison against the anatomical/measurement criterion in Lang et al.; it is not a claim of expert-annotated ground-truth accuracy.

4. Overall standards-based conclusion

Conclusion from the submitted images: The displayed algorithm output is anatomically consistent with the measurement boundary specified by the 2015 ASE/EACVI chamber-quantification recommendations: the LV cavity is delineated along the blood–tissue interface/endocardial border, rather than by the epicardial outer wall.

Across the seven examples, the yellow contours visually behave as cavity contours: they enclose the blood pool, follow the inner myocardial interface, extend toward the LV apex, and use a basal closing segment. This is the correct anatomical target if the intended output is LV cavity area.

What this proves: that your chosen measurement target is aligned with the ASE/EACVI standard.

What this does not prove: that every pixel of every contour is accurate; that the algorithm has high Dice/IoU; that the area values are clinically accurate; or that the method is equivalent to an expert echocardiographer. Those claims require an independent reference annotation and quantitative validation.

6. References and source trail

1.LangRM, Badano LP, Mor-AviV, AfilaloJ, Armstrong A, Ernande L, et al. Recommendations for Cardiac Chamber

Quantification by Echocardiography in Adults: An Update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. Journal of the American Society of Echocardiography. 2015;28(1):1–39.e14. doi:10.1016/j.echo.2014.10.003.

2. Lang RM, Badano LP, Mor-Avi V, et al. Recommendations for Cardiac Chamber Quantification by Echocardiography in Adults: An Update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. European Heart Journal – Cardiovascular Imaging. 2015;16(3):233–270. doi:10.1093/ehjci/jev014.

Key source locations in the 2015 ASE-hosted PDF: pp. 3–5, section “The Left Ventricle / Measurement of LV Size,” including Table 1. The relevant language identifies LV volumetric tracing with the blood–tissue interface and discusses endocardial border tracings and endocardial delineation.

Official source: American Society of Echocardiography — 2015 Chamber Quantification guideline. The official ASE page provides the guideline and PDF.

Prepared from: the seven-page user-supplied PDF containing the original echocardiographic frames and the corresponding algorithm-generated yellow LV contours.

Final scientific qualification: This document is a standards-based visual validation report. For a publication-grade validation, the next step would be to compare the algorithm contour with an independent reference contour using Dice similarity coefficient/IoU, Hausdorff distance or mean contour distance, and area error, preferably across multiple frames and subjects.